

**S.Y. B. Tech. Academic Year 2025-26 Semester: II**  
**CSE20020: Data Structures Laboratory**

**WRITE UP**

**Experiment Number: 01**

**TITLE: Sorting**

**PROBLEM STATEMENT:**

Write a C program to create a student database using an array of structures. Apply Sorting techniques(Bubble sort ,Insertion Sort, Selection Sort).

**OBJECTIVE:**

1. To understand use of an array of structures for maintaining records
2. To implement and analyze sorting techniques
3. To Compare Bubble sort ,Insertion Sort, Selection Sort.

**IMPLEMENTATION:**

- **PLATFORM:**

- 64-bit Open source Linux or its derivatives.
- Open Source C Programming tool like gcc/Eclipse Editor.

- **TEST CONDITIONS:-**

- 1.Input at least five records.
- 2.Arrange list of students roll call listwise . (Use Bubble sort )
3. Arrange list of students rollcall listwise . (Use Insertion sort and Selection Sort )
- 4.Test three sorting methods for at least five records.

- **PSEUDO CODE:**      *//To be Written by Students*

Write pseudocode Bubble, Selection and Insertion sort.

- **TIME COMPLEXITY:**                *//To be Written by Students*

Find out time complexity of above operations

- **CONCLUSION:**

Thus, implemented different sorting methods on the student database. This System is able to perform sorting under different cases.

- **FAQs**                                *//To be Written by Students*

1. What is the meaning of database ? How to maintain in C
2. What are the applications of sorting ?
3. Compare and contrast bubble,selection,insertion sort and shell sort?
4. What is external & internal sorting ,In-place ,Stable Features of Sorting Algorithms?

- **PRACTICE ASSIGNMENTS**

1. Write a program to insert a new record into existing record
2. Write a program to delete a record from existing record
3. Write a program for bubble sort whose best case time complexity is  $O(n)$  .
4. Write a program to arrange list of students in alphabetical order (Use Selection Sort )